

Unit 2 — Scientific Notation

★ Answer Key ★

Do not distribute to students until after the lesson.

Guided Notes — Fill-Ins

2.2 Scientific Notation. Large → +exp; Small → -exp. Multiply: MULTIPLY coefficients, ADD exponents, adjust to 1-10. Divide: DIVIDE coefficients, SUBTRACT exponents. Examples: $(3 \times 10^4)(2 \times 10^5) = 6 \times 10^9$; $(8 \times 10^6) \div (4 \times 10^2) = 2 \times 10^4$; $(5 \times 10^3)(4 \times 10^2) = 20 \times 10^5 \rightarrow 2.0 \times 10^6$.

Practice A — Conversions

1) 4.5×10^7 2) 3.2×10^{-4} 3) 602,000,000 4) 0.000091 5) 7.8×10^5 6) 6.7×10^{-6} 7) 34,500 8) 0.0012 9) 5.28×10^9 10) 0.025 11) 9.04×10^{-7} 12) 7,150,000 13) 6.34×10^{10} 14) 0.0000088

Practice B — Operations

1) 6×10^9 2) 2×10^4 3) 3.5×10^{10} 4) 3×10^3 5) 3.0×10^{10} 6) 6×10^4 7) 4×10^{11} 8) 2.0×10^4

Practice C — Word Problems

1) $(9.3 \times 10^7) \div (1.86 \times 10^5) = 5 \times 10^2$ s (500 s) 2) 40× longer 3) $1 \times 10^5 = \$100,000$ 4) 5.2×10^{15} g 5) 3.7×10^8 m 6) 3×10^2 s (300 s) 7) $(2 \times 10^{12}) \div (4 \times 10^6) = 5 \times 10^5$ photos 8) $(3 \times 10^8)(6 \times 10^3) = 1.8 \times 10^{12}$ m

Cumulative Review

A • Conversions: 1) 7.2×10^6 2) 4.5×10^{-4} 3) 30,500,000 4) 0.000061 5) 9.1×10^4 6) 8×10^{-7} 7) 4,440,000 8) 0.0027

B • Operations: 1) 9×10^9 2) 4×10^6 3) 2×10^{10} 4) 6×10^6 5) 5.6×10^8 6) 4.8×10^2

C • Word Problems: 1) 4× wider 2) 5×10^3 h (5,000 h) 3) 1×10^6 widgets 4) $(6 \times 10^{-5}) \div (3 \times 10^{-7}) = 2 \times 10^2 = 200 \times$ larger